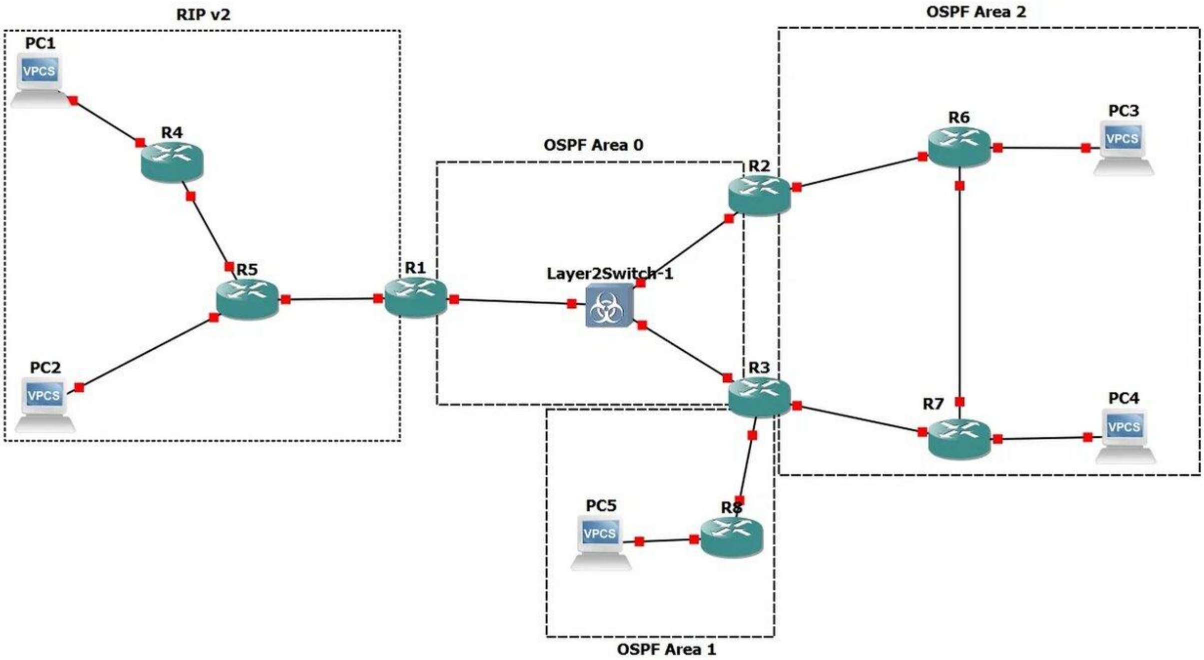
Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF




1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

| Сеть | Подсеть | Устройства |
|---|---|---|
| PC1-R4 | 192.168.1.0/24 | PC1: .1<br>R4: .2 |
| PC2-R5 | 192.168.2.0/24 | PC2: .1<br>R5: .2 |
| R4-R5 | 10.0.12.0/24 | R4: .1<br>R5: .2 |
| R5-R1 | 10.0.13.0/24 | R5: .1<br>R1: .2 |
| R1-SW1-R2-R3 | 10.0.0.0/24 | R1: .1<br>R2: .2<br>R3: .3 |
| R2-R6 | 172.16.1.0/24 | R2: .1<br>R6: .2 |
| R6-PC3 | 172.16.2.0/24 | R6: .1<br>PC3: .2 |
| R3-R7 | 172.16.3.0/24 | R3: .1<br>R7: .2 |
| R7-PC4 | 172.16.4.0/24 | R7: .1<br>PC4: .2 |
| R6-R7 | 172.16.5.0/24 | R6: .1<br>R7: .2 |
| R3-R8 | 192.168.21.0/24 | R3: .1 R8: .2 |

| | | |
|---|---|---|
| R8-PC5 | 192.168.22.0/24 | R8: .1<br>PC5: .2 |

R1:

conf t

interface FastEthernet0/0

ip address 10.0.13.2 255.255.255.0

no shutdown

exit

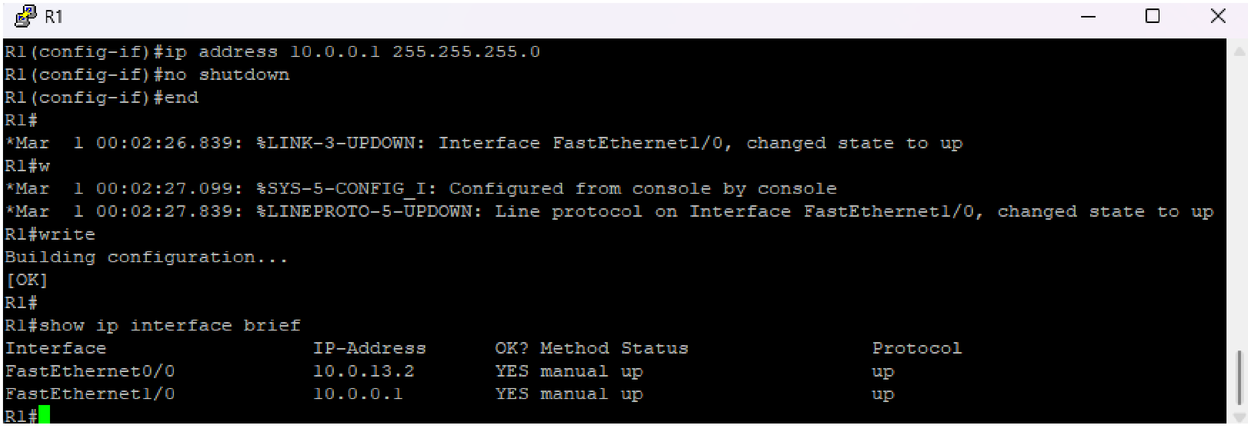
interface FastEthernet1/0

ip address 10.0.0.1 255.255.255.0

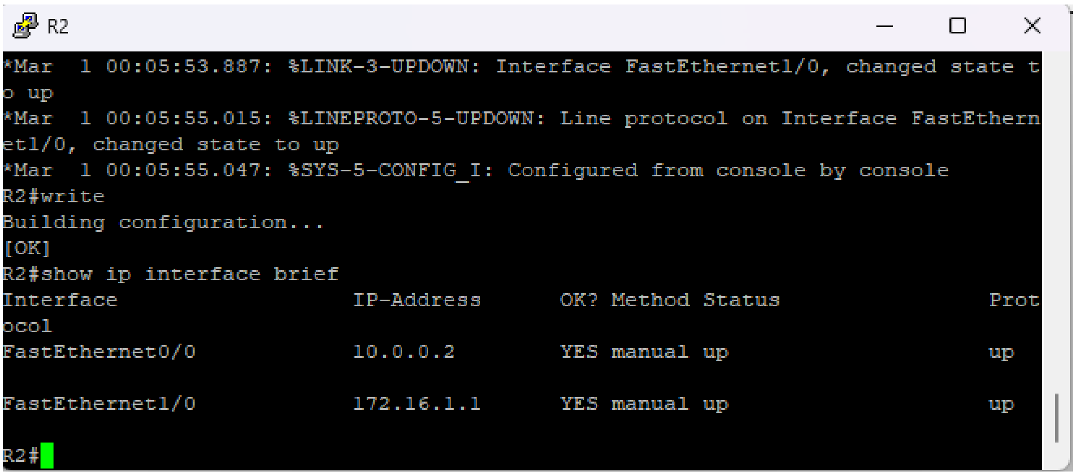
no shutdown

end

write




Настройка аналогично на других маршрутизаторах



R3

```
R3(config)#interface FastEthernet2/0
R3(config-if)#ip address 192.168.21.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
*Mar  1 00:08:41.067: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Mar  1 00:08:42.067: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R3(config-if)#end
R3#wri
*Mar  1 00:08:45.607: %SYS-5-CONFIG_I: Configured from console by console
R3#write
Building configuration...
[OK]
R3#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.0.0.3        YES manual up                    up
FastEthernet1/0            172.16.3.1      YES manual up                    up
FastEthernet2/0            192.168.21.1    YES manual up                    up
R3#
```

R4

```
et1/0, changed state to up
R4(config-if)#end
R4#w
*Mar  1 00:09:03.707: %SYS-5-CONFIG_I: Configured from console by console
R4#write
Building configuration...
[OK]
R4#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            192.168.1.2     YES manual up                    up

FastEthernet1/0            10.0.12.1       YES manual up                    up

FastEthernet2/0            unassigned      YES unset  administratively down down

R4#
```

R5

```
R5#wr
*Mar  1 00:11:06.959: %SYS-5-CONFIG_I: Configured from console by console
R5#write
Building configuration...
[OK]
R5#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            192.168.2.2     YES manual up                    up

FastEthernet1/0            10.0.12.2       YES manual up                    up

FastEthernet2/0            10.0.13.1       YES manual up                    up

R5#
```

R6

```
et2/0, changed state to up
R6(config)#end
R6#w
*Mar  1 00:12:35.299: %SYS-5-CONFIG_I: Configured from console by console
R6#write
Building configuration...
[OK]
R6#show ip interface brief
Interface                  IP-Address      OK? Method Status                Pr
ocol
FastEthernet0/0            172.16.1.2      YES manual up                    up

FastEthernet1/0            172.16.2.1      YES manual up                    up

FastEthernet2/0            172.16.5.1      YES manual up                    up

R6#
```

R7

```
*Mar  1 00:13:40.051: %SYS-5-CONFIG_I: Configured from console by console
R7#write
Building configuration...
[OK]
R7#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            172.16.3.2      YES manual up                    up

FastEthernet1/0            172.16.4.1      YES manual up                    up

FastEthernet2/0            172.16.5.2      YES manual up                    up

R7#
```

R8

```
R8#w
*Mar  1 00:14:28.475: %SYS-5-CONFIG_I: Configured from console by console
R8#write
Building configuration...
[OK]
R8#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            192.168.21.2    YES manual up                    up

FastEthernet1/0            192.168.22.1    YES manual up                    up

R8#
```

PC1 startup.vpc

```
# This the configuration for PC1
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 192.168.1.1 255.255.255.0 192.168.1.2
#

set pcname PC1
```

PC2 startup.vpc

```
# This the configuration for PC2
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 192.168.2.1 255.255.255.0 192.168.2.2
#

set pcname PC2
```

PC3 startup.vpc

```
# This the configuration for PC3
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 172.16.2.2 255.255.255.0 172.16.2.1
#

set pcname PC3
```

PC4 startup.vpc

```
# This the configuration for PC4
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 172.16.4.2 255.255.255.0 172.16.4.1
#

set pcname PC4
```

PC5 startup.vpc

```
# This the configuration for PC5
#
# Uncomment the following line to enable DHCP
# dhcp
# or the line below to manually setup an IP address and subnet mask
ip 192.168.22.2 255.255.255.0 192.168.22.1
#

set pcname PC5
```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

**R4:**

```
conf t
router rip
version 2
network 192.168.1.0
network 10.0.12.0
no auto-summary
end
write
```

**R5:**

```
conf t
router rip
version 2
network 192.168.2.0
network 10.0.12.0
network 10.0.13.0
no auto-summary
end
write
```

**R1:**

```
conf t
```

router rip

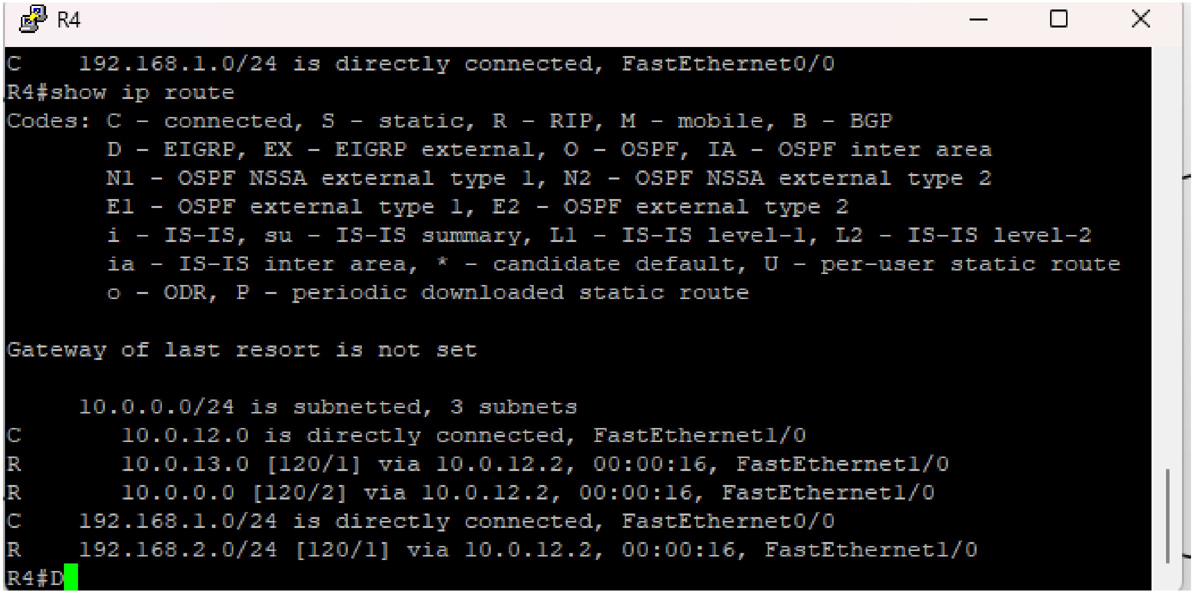
version 2

network 10.0.13.0

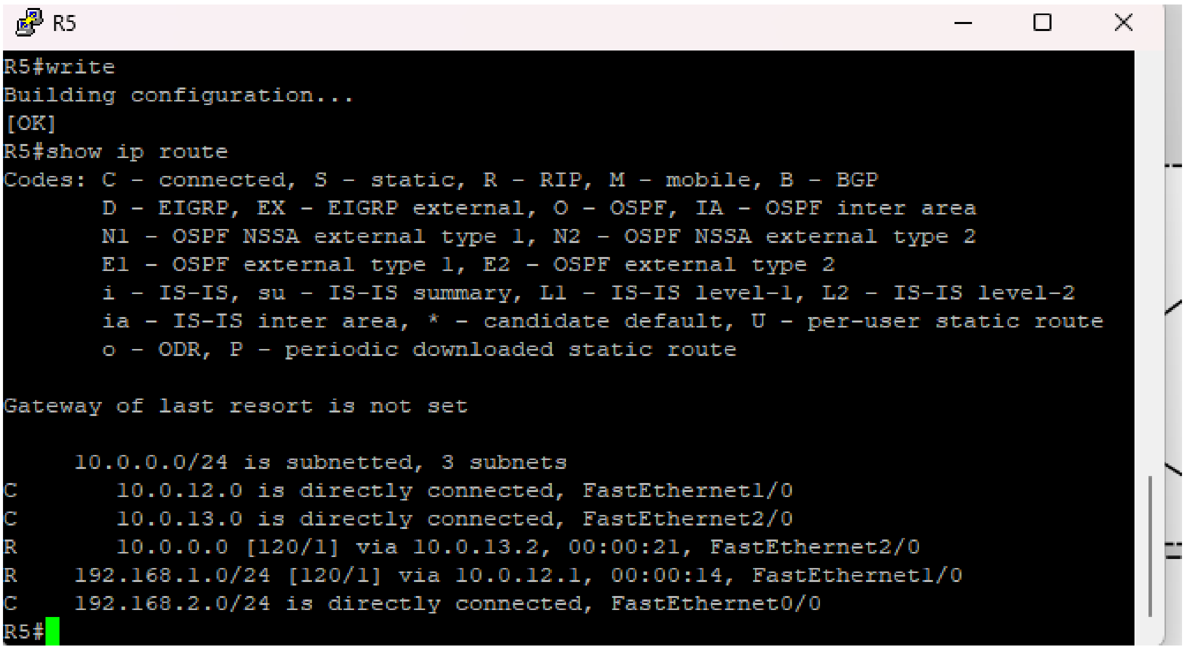
no auto-summary

end

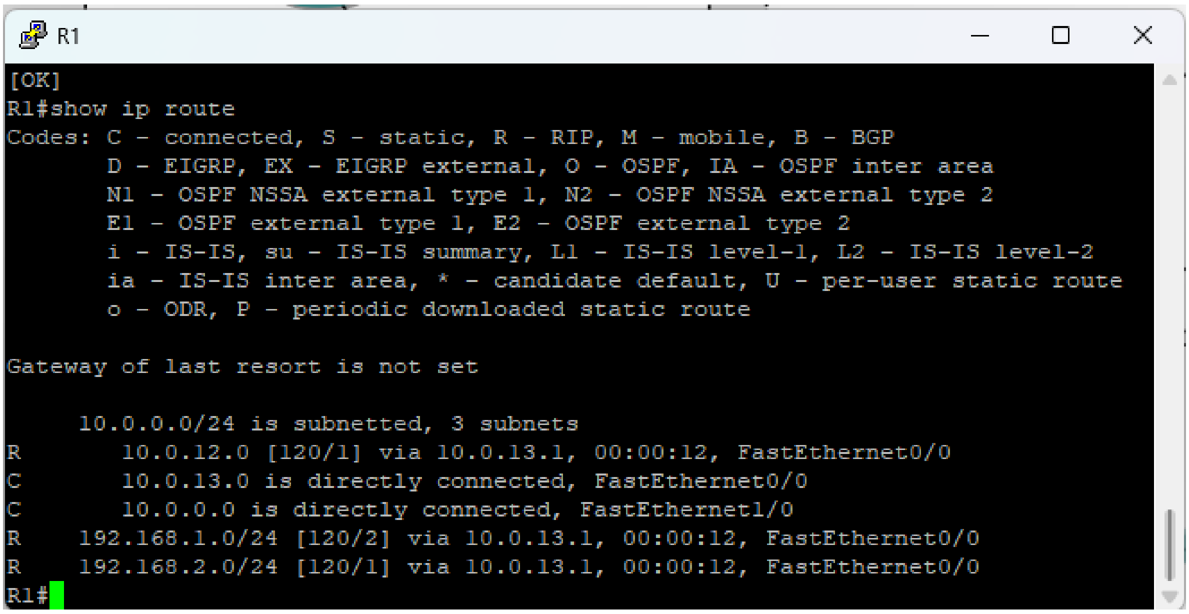
write



```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.0.12.0 is directly connected, FastEthernet1/0
R       10.0.13.0 [120/1] via 10.0.12.2, 00:00:16, FastEthernet1/0
R       10.0.0.0 [120/2] via 10.0.12.2, 00:00:16, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.12.2, 00:00:16, FastEthernet1/0
R4#D
```



```
R5#write
Building configuration...
[OK]
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
C       10.0.12.0 is directly connected, FastEthernet1/0
C       10.0.13.0 is directly connected, FastEthernet2/0
R       10.0.0.0 [120/1] via 10.0.13.2, 00:00:21, FastEthernet2/0
R    192.168.1.0/24 [120/1] via 10.0.12.1, 00:00:14, FastEthernet1/0
C    192.168.2.0/24 is directly connected, FastEthernet0/0
R5#
```




```
[OK]
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 3 subnets
R       10.0.12.0 [120/1] via 10.0.13.1, 00:00:12, FastEthernet0/0
C       10.0.13.0 is directly connected, FastEthernet0/0
C       10.0.0.0 is directly connected, FastEthernet1/0
R    192.168.1.0/24 [120/2] via 10.0.13.1, 00:00:12, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.13.1, 00:00:12, FastEthernet0/0
R1#
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

**R1:**

conf t

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

end

write

**R2:**

conf t

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

network 172.16.1.0 0.0.0.255 area 2

end

write

**R3:**

conf t

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

network 172.16.3.0 0.0.0.255 area 2

network 192.168.21.0 0.0.0.255 area 1

end

write

**R6:**

conf t

router ospf 1

network 172.16.1.0 0.0.0.255 area 2

network 172.16.2.0 0.0.0.255 area 2

network 172.16.5.0 0.0.0.255 area 2

end

write

**R7:**

conf t

router ospf 1

network 172.16.3.0 0.0.0.255 area 2

network 172.16.4.0 0.0.0.255 area 2

network 172.16.5.0 0.0.0.255 area 2

end

write

**R8:**

conf t

router ospf 1

area 1 stub no-summary

network 192.168.21.0 0.0.0.255 area 1

network 192.168.22.0 0.0.0.255 area 1

end

write

**R3:**

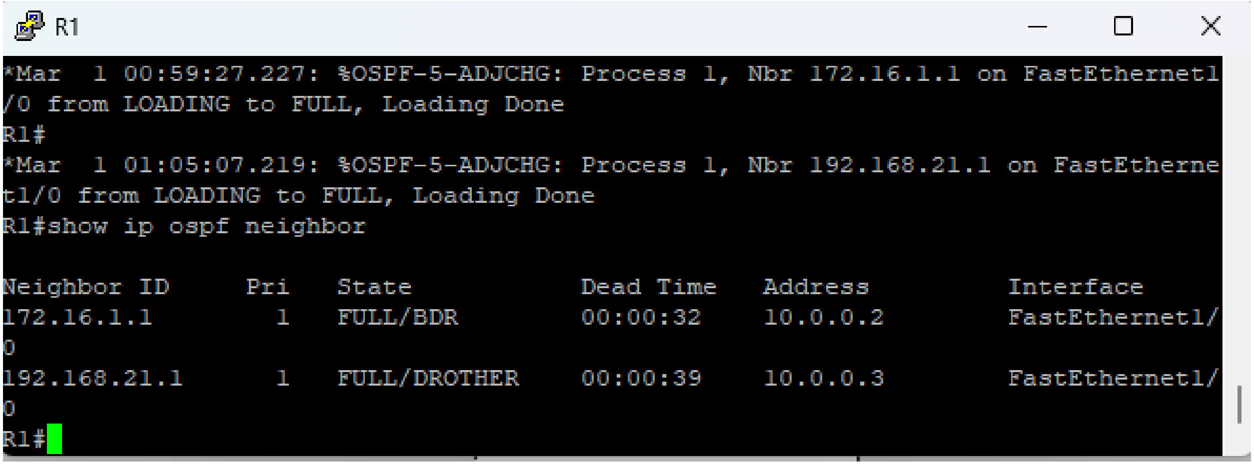
conf t

router ospf 1

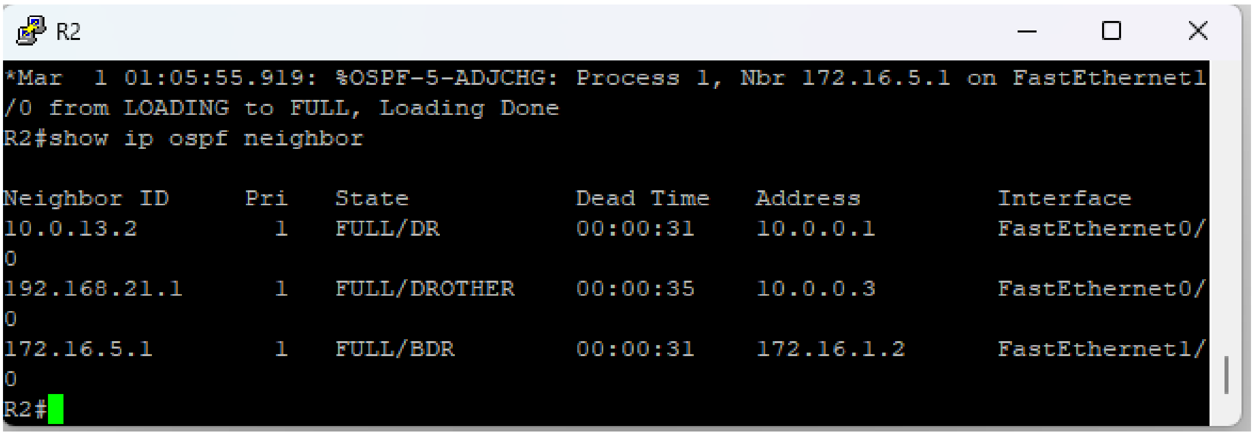
area 1 stub no-summary

end

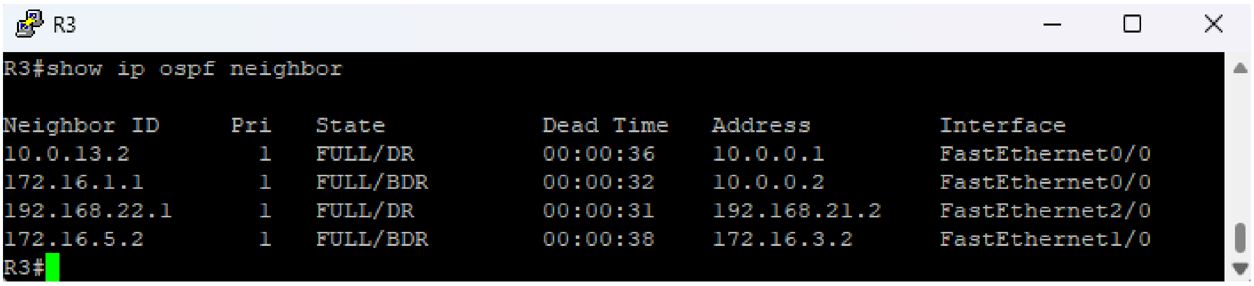
write



```
R1
*Mar  1 00:59:27.227: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.1.1 on FastEthernet1
/0 from LOADING to FULL, Loading Done
R1#
*Mar  1 01:05:07.219: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.21.1 on FastEtherne
t1/0 from LOADING to FULL, Loading Done
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.1.1        1   FULL/BDR        00:00:32    10.0.0.2        FastEthernet1/
0
192.168.21.1      1   FULL/DROTHER    00:00:39    10.0.0.3        FastEthernet1/
0
R1#
```



```
R2
*Mar  1 01:05:55.919: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.5.1 on FastEthernet1
/0 from LOADING to FULL, Loading Done
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.13.2         1   FULL/DR         00:00:31    10.0.0.1        FastEthernet0/
0
192.168.21.1      1   FULL/DROTHER    00:00:35    10.0.0.3        FastEthernet0/
0
172.16.5.1        1   FULL/BDR        00:00:31    172.16.1.2      FastEthernet1/
0
R2#
```



```
R3
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.13.2         1   FULL/DR         00:00:36    10.0.0.1        FastEthernet0/0
172.16.1.1        1   FULL/BDR        00:00:32    10.0.0.2        FastEthernet0/0
192.168.22.1      1   FULL/DR         00:00:31    192.168.21.2    FastEthernet2/0
172.16.5.2        1   FULL/BDR        00:00:38    172.16.3.2      FastEthernet1/0
R3#
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

conf t
router ospf 1
redistribute rip subnets
router rip
redistribute ospf 1 metric 4
end
write

R1

```
[OK]
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 5 subnets
O IA    172.16.4.0 [110/3] via 10.0.0.3, 00:00:52, FastEthernet1/0
O IA    172.16.5.0 [110/3] via 10.0.0.3, 00:00:52, FastEthernet1/0
                   [110/3] via 10.0.0.2, 00:00:52, FastEthernet1/0
O IA    172.16.1.0 [110/2] via 10.0.0.2, 00:00:52, FastEthernet1/0
O IA    172.16.2.0 [110/3] via 10.0.0.2, 00:00:52, FastEthernet1/0
O IA    172.16.3.0 [110/2] via 10.0.0.3, 00:00:52, FastEthernet1/0
O IA 192.168.21.0/24 [110/2] via 10.0.0.3, 00:00:52, FastEthernet1/0
     10.0.0.0/24 is subnetted, 3 subnets
R       10.0.12.0 [120/1] via 10.0.13.1, 00:00:09, FastEthernet0/0
C       10.0.13.0 is directly connected, FastEthernet0/0
C       10.0.0.0 is directly connected, FastEthernet1/0
O IA 192.168.22.0/24 [110/3] via 10.0.0.3, 00:00:53, FastEthernet1/0
R    192.168.1.0/24 [120/2] via 10.0.13.1, 00:00:17, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.13.1, 00:00:17, FastEthernet0/0
R1#
R1#
R1#
R1#
R1#
```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми".

```
PC1> ping 192.168.2.1

192.168.2.1 icmp_seq=1 timeout
84 bytes from 192.168.2.1 icmp_seq=2 ttl=62 time=20.402 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=62 time=25.223 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=62 time=25.630 ms
^C
PC1> ping 172.16.2.2

172.16.2.2 icmp_seq=1 timeout
84 bytes from 172.16.2.2 icmp_seq=2 ttl=59 time=56.453 ms
84 bytes from 172.16.2.2 icmp_seq=3 ttl=59 time=55.397 ms
84 bytes from 172.16.2.2 icmp_seq=4 ttl=59 time=55.474 ms
^C
PC1> ping 172.16.4.2

172.16.4.2 icmp_seq=1 timeout
84 bytes from 172.16.4.2 icmp_seq=2 ttl=59 time=56.594 ms
84 bytes from 172.16.4.2 icmp_seq=3 ttl=59 time=55.684 ms
84 bytes from 172.16.4.2 icmp_seq=4 ttl=59 time=55.543 ms
^C
PC1> ping 192.168.22.2

192.168.22.2 icmp_seq=1 timeout
84 bytes from 192.168.22.2 icmp_seq=2 ttl=59 time=55.139 ms
84 bytes from 192.168.22.2 icmp_seq=3 ttl=59 time=54.783 ms
^C
PC1>
```

```
PC2> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=62 time=39.651 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=62 time=25.454 ms
^C
PC2> ping 172.16.2.2

84 bytes from 172.16.2.2 icmp_seq=1 ttl=60 time=46.517 ms
84 bytes from 172.16.2.2 icmp_seq=2 ttl=60 time=45.888 ms
^C
PC2> ping 172.16.4.2

84 bytes from 172.16.4.2 icmp_seq=1 ttl=60 time=41.912 ms
84 bytes from 172.16.4.2 icmp_seq=2 ttl=60 time=55.995 ms
^C
PC2> ping 192.168.22.2

84 bytes from 192.168.22.2 icmp_seq=1 ttl=60 time=49.097 ms
84 bytes from 192.168.22.2 icmp_seq=2 ttl=60 time=55.515 ms
84 bytes from 192.168.22.2 icmp_seq=3 ttl=60 time=45.505 ms
^C
PC2>
```

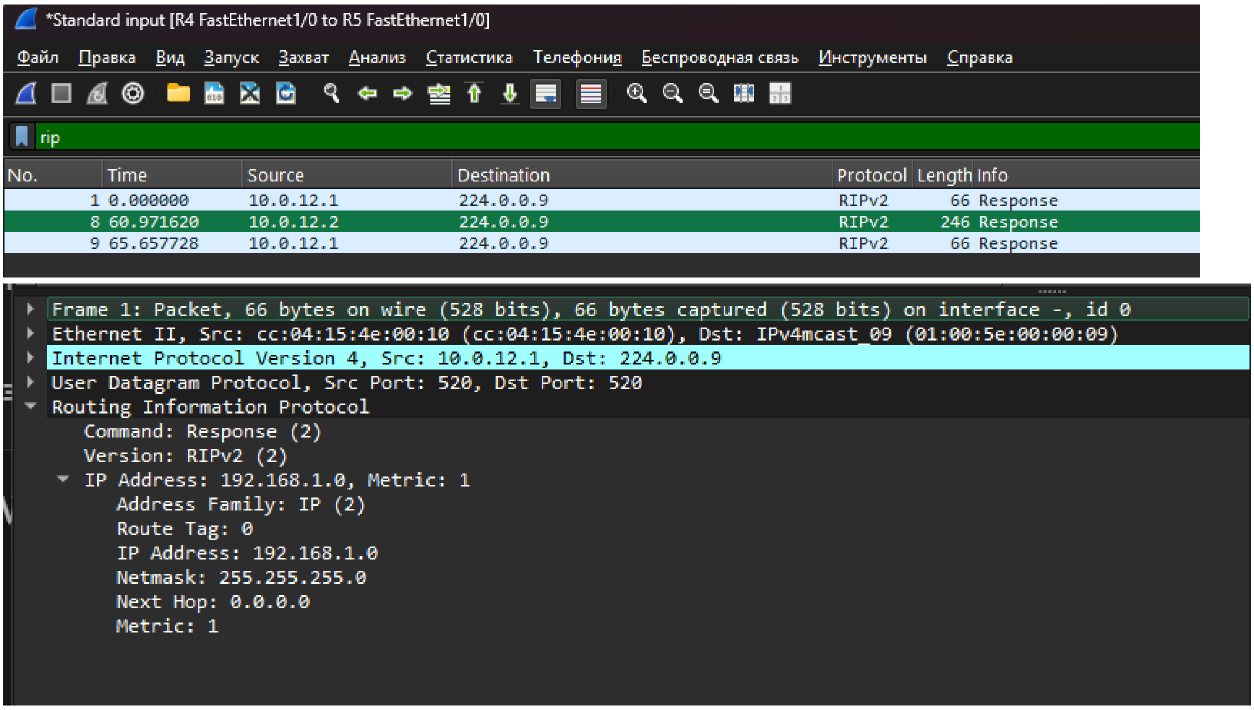
```
PC3>
PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=59 time=70.166 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=59 time=65.522 ms
^C
PC3> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=60 time=51.486 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=60 time=45.958 ms
^C
PC3> ping 172.16.4.2

84 bytes from 172.16.4.2 icmp_seq=1 ttl=62 time=39.319 ms
84 bytes from 172.16.4.2 icmp_seq=2 ttl=62 time=25.238 ms
^C
PC3> ping 192.168.22.2

84 bytes from 192.168.22.2 icmp_seq=1 ttl=60 time=56.159 ms
84 bytes from 192.168.22.2 icmp_seq=2 ttl=60 time=45.941 ms
^C
PC3>
```

```
PC4>
PC4> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=59 time=78.671 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=59 time=65.595 ms
^C
PC4> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=60 time=54.484 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=60 time=55.513 ms
^C
PC4> ping 172.16.2.2

84 bytes from 172.16.2.2 icmp_seq=1 ttl=62 time=37.458 ms
84 bytes from 172.16.2.2 icmp_seq=2 ttl=62 time=25.247 ms
^C
PC4> ping 192.168.22.2

84 bytes from 192.168.22.2 icmp_seq=1 ttl=61 time=40.339 ms
84 bytes from 192.168.22.2 icmp_seq=2 ttl=61 time=35.115 ms
^C
PC4>
```

```
PC5>
PC5> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=59 time=59.524 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=59 time=55.622 ms
^C
PC5> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=60 time=63.751 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=60 time=55.366 ms
^C
PC5> ping 172.16.2.2

84 bytes from 172.16.2.2 icmp_seq=1 ttl=60 time=55.112 ms
84 bytes from 172.16.2.2 icmp_seq=2 ttl=60 time=56.323 ms
^C
PC5> ping 172.16.4.2

84 bytes from 172.16.4.2 icmp_seq=1 ttl=61 time=44.914 ms
84 bytes from 172.16.4.2 icmp_seq=2 ttl=61 time=35.516 ms
84 bytes from 172.16.4.2 icmp_seq=3 ttl=61 time=35.782 ms
^C
PC5>
```

6) Перехватить в wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

RIP v2:




Src: 10.0.12.1 (R4)

Dst: 224.0.0.9 - multicast адрес RIP v2

Command: Response (2) - роутер рассылает свою таблицу маршрутизации

Version: RIPv2 (2) - версия 2

IP Address: 192.168.1.0 - анонсируемая сеть (сеть PC1-R4)

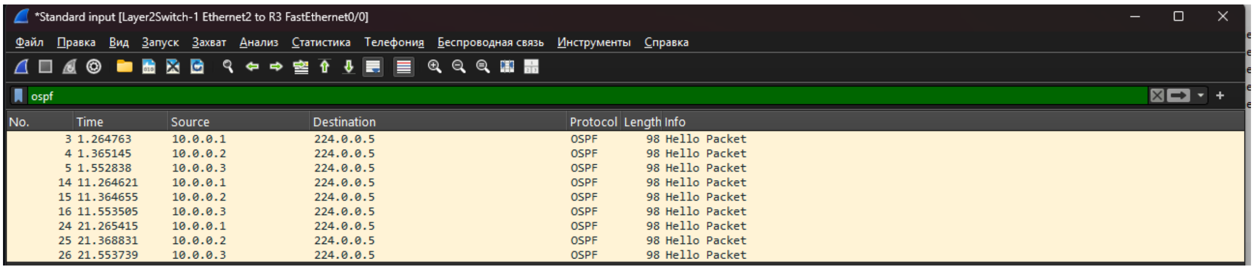
Netmask: 255.255.255.0 - маска

Next Hop: 0.0.0.0 - следующий хоп = сам отправитель

Metric: 1 - 1 хоп до этой сети

Route Tag: 0 - метка маршрута (используется при редистрибуции)

OSPF:



```
▸ Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▸ Ethernet II, Src: cc:02:15:12:00:00 (cc:02:15:12:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
▸ Internet Protocol Version 4, Src: 10.0.0.2, Dst: 224.0.0.5
▾ Open Shortest Path First
  ▾ OSPF Header
      Version: 2
      Message Type: Hello Packet (1)
      Packet Length: 52
      Source OSPF Router: 172.16.1.1
      Area ID: 0.0.0.0 (Backbone)
      Checksum: 0x3ed5 [correct]
      Instance ID: Base IPv4 Unicast Instance (0)
      Auth Type: Null (0)
      Auth Data (none): 0000000000000000
  ▾ OSPF Hello Packet
      Network Mask: 255.255.255.0
      Hello Interval [sec]: 10
    ▸ Options: 0x12, (L) LLS Data block, (E) External Routing
      Router Priority: 1
      Router Dead Interval [sec]: 40
      Designated Router: 10.0.0.3
      Backup Designated Router: 10.0.0.1
      Active Neighbor: 10.0.13.2
      Active Neighbor: 192.168.21.1
  ▾ OSPF LLS Data Block
      Checksum: 0xfff6
      LLS Data Length: 12 bytes
    ▸ Extended options TLV
```

Dst: 224.0.0.5 - multicast (для всех OSPF-роутеров)

OSPF Header:

Version: 2 - OSPFv2

Message Type: Hello Packet (1) - тип пакета — поиск и поддержание соседей

Source OSPF Router: 172.16.1.1 - Router ID отправителя (R2)

Area ID: 0.0.0.0 (Backbone) - пакет Area 0

OSPF Hello Packet:

Network Mask: 255.255.255.0 - маска интерфейса

Hello Interval: 10 - как часто шлёт Hello

Router Dead Interval: 40 - если 40 сек нет Hello, значит сосед мёртв

Router Priority: 1 - приоритет для выбора DR/BDR

Designated Router: 10.0.0.3 = R3

Backup DR: 10.0.0.1 - BDR = R1

Active Neighbor: 10.0.13.2 - известные соседи

Active Neighbor: 192.168.21.1